Health Insurance Fraud Detection Using Feature Selection and Ensemble Machine Learning Techniques

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Abstract Frauds existing in the healthcare system have led to additional expenses as well as caused degradation of the quality of services being provided to patients. In this research, we have proposed a Sequential Forward Selection (SFS) method and SMOTE oversampling approach to healthcare insurance fraud detection. We employed K-Nearest Neighbors (KNN), Artificial Neural Network (ANN), Linear Discriminant Analysis (LDA), Gradient Boosting Machine (GBM), Bagging classifier, and stacking meta-estimator for the classification purpose. We preferred Stacking aggregator over others in fusion with SFS as there is a statistically significant difference between them. This work categorizes, compares, and summarizes almost all published technical and review articles in healthcare fraud detection over the past 10 years. It defines the professional fraudster, formalizes the main types and subtypes of known fraud, and presents the nature of data evidence collected within affected industries. Within the business context of mining the data to achieve higher cost savings, this research presents methods and techniques together with their problems. The accuracy is best obtained when feature selection is applied to the Stacking classifier which is 97.19%. To demonstrate the effectiveness of the proposed methodology, we have experimentally evaluated our proposition using a real-world healthcare insurance fraud dataset taken from the literature.

Keywords SMOTE · Sequential forward selection · K-nearest neighbor · Artificial neural network · Adaboost · Linear discriminant analysis · Gradient boosting machine · Bagging · Stacking

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1 Introduction

Fraud can be defined as anything wrong or having a criminal deception which leads to financial loss or personal gain to an entity. According to the National Health Care Anti-Fraud Association, health care fraud is an intentional craftiness made by a person, or an entity that could result in some unauthorized benefits to him or her accomplices [1]. Fraud causes difficulties in our day-to-day activities and also at times incurs great losses to insurance companies. Healthcare has emerged as one of the important aspects of human lives, but at the same time, it involves high expenditure which abstains the patients from taking up the services. Many government-funded healthcare insurance policies rent out money to the needy and reimburse the amount at a later period.

Due to healthcare abuse and fraud, a country's economy faces a huge financial loss. It is estimated that 10% of medical system expenditure is wasted due to fraud, making it an alarming concern. A report suggests that the health care industry in India is losing upto ₹600–₹800 crores on fraudulent claims annually [2]. A safe and reliable environment should be built for efficient fraud detection and for this purpose, the very step is the identification of the fraud and then applying suitable approaches to detect and eradicate the bug from the system should be established.

The healthcare fraud detection can be either anomaly-based or misuse-based detection system where is the first approach, the model is trained with normal claims profiles and any deviation from the normal behavior is labeled as intrusive. On the contrary, in the misuse-based detection scenario, a misuse signature is built and any incident claim when matched with the misuse pattern is flagged as a fraud. Data mining is an efficient tool which helps in exploring new variants of the useful information from a huge raw data. It helps in decision making. Computerized statistical tools are used with integration with data mining approach. The two distinct analysis tools that can be enumerated are- supervised and unsupervised learning. The supervised learning works on labeled datasets whereas the unsupervised approach is applied to unlabeled datasets. Another improvisation of machine learning is the application of ensemble methodologies to several base learners to develop a final efficient predictor. Instead of repeatedly executing the base models to improve the performance ensemble methods can be used to boost up the performance.

Recent times have witnessed health insurance fraud as a serious threat. Health insurance companies lose an enormous money as a part of these fraudulent activities. Even service providers and at times patients also manipulate the policies and carry out fraud to gain unauthorized financial benefits. Hence, healthcare insurance detection has become crucial. Such scenarios at times cause harassment to legitimate clients and does not allow the company to deliver the required sum of money to the needy patients at the right time. Hence, lowering the effectiveness and working of the health insurance companies.

The main goal of the research is to devise an efficient model that could detect health insurance fraud, that is it should be able to detect fraudulent claims accurately, attempt to increase the performance metrics and minimizing the misclassification rate. It would basically help to improvise the existing health insurance company's system to detect anomalies accurately and in less time. According to the current worldwide economy, an efficient insurance company is one that fulfills its customer's needs and demands and safeguards its patient's documents without any ambiguity. Most insurance companies use obsolete techniques to find abusive claims and this makes way for frauds to creep into the system. The primary task of a health insurance company is to identify the fraud claims and restore the losses incurred by the customer. It is obligatory to devise a system to trace the customer's records which would reduce fraud happening in the system and deliver quality health care services.

1.1 Categories of Health Insurance Frauds

The frauds which are encountered in the healthcare industries can be under three categories. Firstly, the one which is carried out by service providers which includes billing services which have not been done, unbundling i.e., billing more costly services which are not required, upcoding which to bill costly services which were not part of the treatment and performing unnecessary medical services which are not needed. In the second place, come the frauds perpetuated by insurance subscribers that is the customers by deceiving the eligibility or employment criteria in order to gain premiums at lower rate, proclaiming medical claims which are actually not received and at time identity theft is also done. The third category is insurance carrier fraud where the insurance companies forge reimbursements, rewards and service statements.

1.2 Our Contributions

The significant contribution of the paper illustrates:

1. Investigation of existing data mining and machine learning methodologies of Artificial Neural Network (ANN), Linear Discriminant Analysis (LDA), Gradient Boosting Machine (GBM), Adaptive Boosting (Adaboost), K-Nearest Neighbor (KNN), Bagging Classifier and Stacking Classifier;
2. Designing and developing the efficient feature subset through Sequential Feature Selection (SFS) technique and then applying the learners and ensembles
3. And then comparing the results of all the classifiers and finding out the best fit for our approach and the dataset.

The remaining work in this paper is organized as follows; Sect. 2 has the literature review of various research papers on the health insurance fraud detection domain, Sect. 3 discusses the methodologies used, Sect. 4 enumerates the experimental analysis and Sect. 5 says about the conclusion and future work.

2 Related Work

In the literature, various methodologies have been proposed to detect fraud in the health insurance domain. Richard A. Bauder and Taghi M. Khoshgoftaar propped a novel work where they drew a comparison between supervised, unsupervised and hybrid learners on 2015 Medicare Provider Utilization Payment Dataset. A total of 10 techniques were used and results out that the hybrid model having the combination of unsupervised autoencoder and supervised neural network shows better accuracy [3]. An approach enumerated by Rawte and Srinivas [4] drew the advantages of both techniques of supervised and unsupervised techniques and incorporated them into a hybrid approach where Evolving Clustering Method (ECM) is used to cluster dynamic data and a Support Vector Machine is used for training and classification. A work by Ayushi Verma, Anu Taneja and Anuja Arora was designed by integrating Association rule mining, clustering and outlier detection where the claims were divided into period-based claims and disease-based claims and it was successfully finding out claims from the existing data [5]. D. Thornton and et al. created a multi-dimensional data model for classifying the frauds based on the seven-level proposed by Sparrow [6]. The seven levels proposed by Sparrow encompasses several frauds like upcoding, unbundling, phantom billing, duplicate billing, bill padding, excessive or unnecessary services and kickbacks. The work done by Wan-Shiou Yang, San-Yih Hwang is a process mining framework where clinical pathways is used and a sensitivity of 64% and specificity of 67% [7].

Pallavi Pandey et al. proposed a detection and analyses approach where logistic regression, decision tree and neural network where the neural network accuracy is 99% [8]. A big data fraud detection using supervised learning approaches where logistic regression was found to be better by given by Matthew Harland et al. [9]. Roshan et al. proposed the Markov model and improvised Markov model with gradient boosting which improved Markov model by 97.10% when applied on health insurance [10]. The novel work proposed by Namrata Ghuse, Pranli Pawar and Amol Potgantwar [11] deep learning algorithms are applied in the form of neural networks as part of unsupervised learning vs the supervised learning of logistics regression and random forest. The accuracy of random forest outperformed. Ananthi Sheshasaayee and Surya Susan Thomas used health care data to which supervised learners of naïve bayes, linear regression, decision trees, random forest, logistic regression and support vector machine [12]. Melih Kirilidog and Cuneyt Asuk [13] proposed an anomaly detection technique which applied to a data obtained from a Turkish insurance company. Lu et al. have made use of heterogenous graph and graph neural network to devise fraud detection model called MHAMFD which focuses on patients' regular doctor visits [14]. A detailed taxonomy has been discussed by K. Kapadiya et al. where a survey is presented of all health insurance fraud detection mechanisms that are carried out using AI and blockchain enabled systems [15].

3 Proposed Methodology

The dataset [16] is obtained by merging three datasets and the merged dataset had a class imbalance problem which is dealt by using SMOTE oversampling technique to sample the minority fraudulent cases. Then, machine learning models are applied to the dataset without including any feature selection algorithm. Next, the classifiers are operated in a feature selection algorithm and the performances are compared. Feature selection aims at reducing the dimensionality of the dataset by identifying the input variables which are required to establish a strong relationship with the target output variables which helps in improving the performance of the model, improves data compatibility in the model and computational statistics are reduced.

3.1 About the Dataset

The dataset basically focuses on the details of the inpatient, outpatient and beneficiary details of each healthcare provider. It is an amalgamation of three datasets namely, Outpatient, Inpatient and Beneficiary datasets.

3.2 Data Preprocessing

The data quality is although very good, but it is seen that some improvements are required to make it robust and efficient.

The values of beneficiary dataset especially in the ChroniCond_. So, 0 and 1 values are imputed where 0 depicts no chronic conditions and 1 for yes i.e., the patient has chronic conditions. All numeric values are imputed with zero. Further, for improving the efficiency ‘WhetherDead’, ‘AdmitForDays’ and ‘Age’ attributes. After merging, the final dataset has 57 attributes and the shape of the test data is 135392 and that of training dataset is 558211.

3.3 Algorithms and Models used for Model Building

3.3.1 Sequential Feature Selection

Sequential Feature Selection is a wrapper method that enables us to add or remove features to form a feature subset based on feature importance in a greedy fashion. At every stage, the estimator either adds or removes the best feature based on the cross-validation score of the estimator. The technique contains two basic components; an objective criterion which is used to minimize the overall features of the dataset to enhance the performance metric. And a sequential search algorithm which either increases or decreases the candidate feature. It can be of two types; Sequential Forward Selection (SFS) where the features are sequentially added to a null set till the additional feature does not reduce the objective criterion and the second variant is Sequential Backward Selection (SBS) where all the input data is merged and united into a set and goes on removing the feature till the extraction does not hamper the criterion.

3.3.2 Machine Learning Techniques Used for Classification

The supervised learning paradigms are used to distinguish fraud and non-fraud scenarios. The base learners like K-Nearest Neighbor (KNN), Linear Discriminant Analysis (LDA) and Artificial Neural Network (ANN) is used. KNN is a straightforward approach which works on the principle of finding out the k nearest neighbors by using a distance parameter. LDA is a rectilinear model where the number of features in the feature subset are maximized in order to gain a better classification. ANN are basically computational methods that are inspired by the human nervous system and brain which is composed of neurons.

Ensemble techniques of Adaboost, Gradient boost, Bagging and Stacking has been used. Boosting is a type of ensemble learning algorithm which aims to increase the efficiency of weak learners. Adaptive boosting, or AdaBoost classifier commences by adjusting a classifier on the original dataset and then fits the extra copies of the classifier on the same datasets where the incorrect weights are adjusted. In the case of Gradient Boosting Machine, the base learner is typically decision tree but at times regression can also be used. The construction of base learners is achieved in a way that they are maximally correlated to a negative gradient of loss function. Bootstrapping aggregator, or a bagging classifier is an ensemble meta-estimator that fits into a base learning classifier randomly of the original dataset and then used voting or averaging to get the final prediction. Stacking classifier where predictions are made using multiple classifiers namely, level-one classifier and a meta-estimator. The baseline models predict the output of test datasets and the second layer has a meta-classifier or regressor which uses up the predictions of the baseline models.

In the proposed model, a modified dataset is built by removing the unbalanced factor from the original dataset which contained 10% of the minority class which disabled the learners to perform well. An oversampling technique called SMOTE (Synthetic Minority Over-Sampling Technique) is applied to generate samples of the minority cases. The modified dataset is given to the Sequential Forward Selection technique to select features based on importance and go on adding the features until further addition does not affect the objective criterion. It is then randomly sampled to obtain training and testing datasets. The training dataset goes through the machine learning techniques of KNN, LDA, ANN, AdaBoost, GBM, Bagging and Stacking. The testing dataset is evaluated to determine the intrusive cases. Lastly, the performance metrics of the classifiers are obtained to get the best learning technique suitable for our dataset for fraud detection.

The proposed model is implemented on a Windows 10 Education 64-bit operating system with an Intel core i-3-6006U processor and 8gb RAM on Python 3.7 environment.

3.4 Performance Metrics

The learners which are implemented in this paper are compared based upon some performance metrics to uplift the qualities are discussed below;

Accuracy

It is defined as the fraction of all the true outcomes including both true positives and true negatives to the sum of all cases under examination.

$$Accuracy = TP + TN/(TP + TN + FP + FN) \tag{1}$$

Sensitivity

The proportion of correctly predicted positive observations to the observations. It is also known as recall or true positive rate.

$$Sensitivity = TP / (TP + FN) \tag{2}$$

Specificity

True negative rate defined as the proportion of false positive to all the predicted observation.

$$Specificity = FP / (TN + FP) \tag{3}$$

MCC

Mathew's Correlation Coefficient is basically a balanced performance measuring parameter and is also known as phi-coefficient (ϕ) which is the estimate of the association between two classes in machine learning.

$$MCC = (TP * TN - FP * TN)/\sqrt{((TP + FP)(TP + FN)(TN + FP)(TN + FN))} \quad (4)$$

G-mean

G-mean is defined as the squared mean of sensitivity and specificity.

$$G - mean = \sqrt{Sensitivity * Specificity} \quad (5)$$

4 Results and Observations

A comparison is drawn out by evaluating the performance metrics of each classifier by first taking all the features and then by applying SFS. The results are tabulated below (Tables 1 and 2, Fig. 1).

Table 1 Comparison table of K-nearest neighbor, Linear Discriminant Analysis (LDA) and Artificial Neural Network (ANN)

Classifier	Performance metric	Using all features	Using SFS
K-Nearest Neighbor	Accuracy	0.9193	0.9379
	Sensitivity	0.9391	0.9525
	Specificity	0.6082	0.7152
	MCC	0.4453	0.5641
	G-mean	0.7558	0.8254
LDA	Accuracy	0.9261	0.9433
	Sensitivity	0.9412	0.9572
	Specificity	0.6739	0.7988
	MCC	0.4882	0.6203
	G-mean	0.7964	0.8723
ANN	Accuracy	0.9316	0.9366
	Sensitivity	0.9410	0.9508
	Specificity	0.7531	0.7322
	MCC	0.5188	0.5777
	G-mean	0.8418	0.9290

Table 2 Comparison table of Gradient Boosting Machine (GBM), AdaBoost, bagging and stacking classifier

Classifiers	Performance metric	Using all features	Using SFS
GBM	Accuracy	0.9162	0.9283
	Sensitivity	0.9658	0.9638
	Specificity	0.5421	0.6089
	MCC	0.5605	0.5905
	G-mean	0.7236	0.7660
Adaboost	Accuracy	0.9360	0.9440
	Sensitivity	0.9415	0.9535
	Specificity	0.8353	0.7994
	MCC	0.5771	0.6240
	G-mean	0.8868	0.8731
Bagging	Accuracy	0.9421	0.9525
	Sensitivity	0.9509	0.9630
	Specificity	0.8021	0.8175
	MCC	0.6097	0.6946
	G-mean	0.8733	0.8873
Stacking	Accuracy	0.9601	0.9719
	Sensitivity	0.9379	0.9813
	Specificity	0.8482	0.8333
	MCC	0.7302	0.7768
	G-mean	0.8919	0.9043

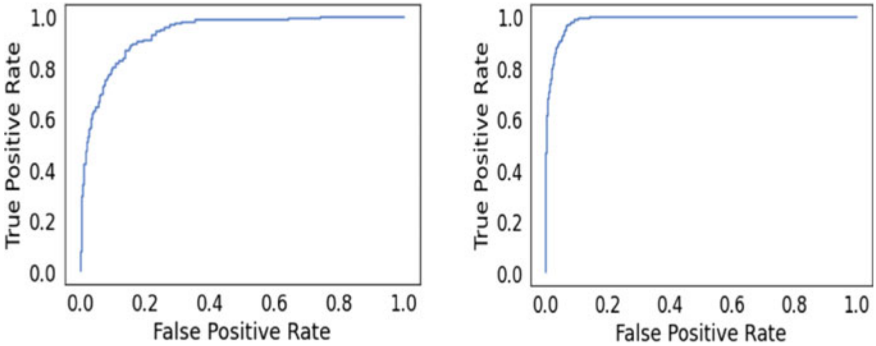

Fig. 1 ROC curve of stacking meta-estimator using all features (left) and using sequential feature selection-SFS (right)

Table 3 Performance evaluation between existing work and proposed approach

	Performance metrics				
	Accuracy	Sensitivity	Specificity	MCC	G-Mean
CatBoost [17]	0.9304	0.9426	0.7191	0.5171	0.8233
Proposed SFS based on stacking classifier	0.9719	0.9813	0.8333	0.7768	0.9043

In the literature, drawing out a comparative analysis between our proposed work and [17], we can deduce the following results (Table 3).

5 Conclusion and Future Scope

The experimental analysis of the mentioned method demonstrates that when feature selection technique is not applied then Stacking meta-estimator outperforms the other classifiers in all the metrics. When Sequential Feature Selection (SFS) technique is applied to the classifiers, then also Stacking meta-estimator shows better performance than the other illustrated classifiers. So, to generalize Stacking generalization classifier when applied to a balanced and feature selected dataset, it shows better performance as compared to other classifiers and takes no iterative loops and less computational time.

The research work has potential future scope if we incorporate optimization technique Differential Evolution and Particle Swarm Optimization to further enhance the performance metrics by applying them to select important features of the dataset and also by reducing the search space. As a part of future prospects, the proposed approach can be applied to various other insurance datasets like automobile, financial, bank, etc. to establish a versatile model.

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